

Roundup[®] Pro Vantage



 - Herbicide

Bayer Amenity



Improved performance



480 g/L glyphosate



Rainfast from 1 hour



Low drift formulation

This product must only be used in England, Scotland and Wales.

For the control of annual and perennial grass and broad-leaved weeds.

A foliar applied translocated herbicide for the control of emerged weeds in industrial and amenity situations, in grassland, orchards, forestry and in aquatic areas.

Degraded by micro-organisms/microbes in the soil.

The (COSHH) Control of Substances Hazardous to Health Regulations may apply to the use of this product at work.

This product contains a soluble concentrate containing 480 g/l glyphosate, present as 588 g/l (43.8% w.w.) of the potassium salt of glyphosate.

MAPP Number 21197

PROTECT FROM FROST

BAYER CROPSCIENCE LIMITED

230 Cambridge Science Park, Milton Road, Cambridge,
CB4 0WB, UK.

Telephone: 01223 226500;

<https://cropscience.bayer.co.uk/>

GROUP

9

HERBICIDE

A1 GB – GB30465150a

e 5 Litres

Crop specific information

CROPS OR SITUATIONS	MAXIMUM INDIVIDUAL DOSE (kg product per hectare)	MAXIMUM TOTAL DOSE (kg product / hectare / crop)	LATEST TIME OF APPLICATION
Permanent grassland (destruction), rotational grassland (destruction).	4.5	4.5 L/ha/year	5 days before harvest, grazing or drilling
Apple and pear orchards.	3.75	3.75 L/ha/year	After harvest (post leaf-fall) but before green cluster stage
Cherry, plum and damson orchards.	3.75	3.75 L/ha/year	After harvest (post leaf-fall) but before white bud stage
All edible and non-edible crops (destruction before sowing/planting)	3.75	3.75 L/ha/year	-
Natural surfaces not intended to bear vegetation, permeable surfaces overlaying soil, hard surfaces	3.75 litres/hectare	-	-
Enclosed waters, open waters, land immediately adjacent to aquatic area.	4.5 litres/hectare	-	-
Forestry: Weed control Stump application	7.5 litres/hectare 150 ml/litre of water (15% solution of product in water)	-	-
Chemical thinning (by injection)	1.5 ml per 10 cm diameter (or less) of tree	-	-
Amenity vegetation	3.75 litres/hectare	-	-
OTHER SPECIFIC RESTRICTIONS: Users must consult the appropriate water regulatory body (Environment Agency/Scottish Environmental Protection Agency/Natural Resources Wales) before using the product near water and must obtain their agreement before using this product to control aquatic weeds. The maximum concentration of active substance in treated water should not exceed 0.2 ppm. When applying through rotary atomisers, the spray droplet spectra produced must be of a minimum Volume Median Diameter (VMD) of 200 microns. Weed wipers may be used in any recommended crop where the wiper or chemical does not touch the growing crop. When using weed wipers, the maximum concentrations used must not exceed the following: Weed wiper Mini 1:3 dilution with water Other Wipers 1:2 dilution with water For stump application the maximum concentration must not exceed 150 ml product per 1.0 litre water.			

AMENITY, INDUSTRIAL AND GENERAL WEED CONTROL

Exclusion Times

People, pets and wildlife need not be kept out of treated areas. It is best not to walk in areas where the spray is still wet as transfer to other vegetation may lead to unwanted damage to other foliage. Once the spray is dry this cannot occur.

Area of Use

Roundup ProVantage is recommended for control of annual and perennial grasses and broad-leaved weeds in non-crop areas, for cleaning up weedy ground prior to planting or sowing and for aquatic weed control. Roundup ProVantage may also be used as a directed spray in ornamental plantings, orchards and for spot treating in grassland. Roundup ProVantage must be targeted only at weed growth on hard surfaces such as roadsides and paths, (see recommendation table for details).

Application Rate

1.0 to 7.5 litres/ha – refer to Recommendation Tables

DIRECTIONS FOR USE

IMPORTANT: This information is approved as part of the Product Label. All instructions within this section must be read carefully in order to obtain safe and successful use of this product.

Warnings

EXTREME CARE SHOULD BE TAKEN TO AVOID SPRAY DRIFT AS THIS CAN SEVERELY DAMAGE NEIGHBOURING CROPS OR PLANTS.

DO NOT MIX, STORE OR APPLY ROUNDUP ProVantage IN GALVANISED OR UNLINED STEEL CONTAINERS OR SPRAY TANKS.

DO NOT leave spray mixtures in tank for long periods and make sure tanks are WELL VENTED.

Restrictions

A period without rain of at least 6 hours and preferably 24 hours must follow application of Roundup ProVantage.

Do not spray onto weeds which are naturally senescent, or where growth is impaired by drought, high temperatures, a covering of dust, flooding or frost at, or immediately after application, otherwise poor control may result.

Do not spray in windy conditions as drift onto desired crops or vegetation can severely damage or destroy them.

Do not tank-mix Roundup ProVantage with adjuvants, pesticides or fertilisers, except as specified in the Compatibility section.

- PERENNIAL GRASSES must have full emergence of healthy, green leaf. (Common Couch, for example, becomes susceptible at the onset of tillering and new rhizome growth, which usually occurs when plants have 4-5 leaves, each with 10-15 cm of new growth).
- PERENNIAL BROAD-LEAVED WEEDS are most susceptible around the flowering stage.
- ANNUAL GRASSES AND BROAD-LEAVED WEEDS should have at least 5 cm of leaf, or 2 expanded true leaves, respectively. In set-aside, annual grasses are best treated at full ear emergence, or before stem elongation. Application during the stem extension phase of annual grasses e.g. Black-grass and Brome species on set-aside between the end of April and end of May, may result in poor control and require re-treatment.

- BRACKEN should be treated after frond tips are unfurled, but pre-senescence.
- OTHER SPECIES – recommendations for specific Areas of Use are given in the Recommendation Tables.
- This product will not give an acceptable level of control of Horsetails (*Equisetum arvense*)-repeat treatment will be necessary.

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H - Herbicide

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Tel: 0330 678 3382 (24 hr)
National Poisons Information Centre UK: 0344 892 0111
(medical professionals only)

ROUNDUP PROVANTAGE UFI: WYU1-TOP0-D0Q2-AADX
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Toxic to aquatic life with long lasting effects.
Wear protective gloves/protective clothing/eye protection/face protection.
Collect spillage.
Dispose of contents/container to a licensed hazardous-waste disposal contractor or collection site except for empty clean containers which can be disposed of as non-hazardous waste.
To avoid risks to human health and the environment, comply with the instructions for use.



IMPORTANT INFORMATION:
FOR PROFESSIONAL USE ONLY AS AN AGRICULTURAL / HORTICULTURAL / INDUSTRIAL / FORESTRY / AQUATIC HERBICIDE
Crops/situations: All edible and non-edible crops (destruction, before sowing/planting). Grassland. Natural surfaces not intended to bear vegetation; permeable surfaces overlaying soil; hard surfaces. Enclosed waters, open waters, land immediately adjacent to aquatic areas. Forest, forest nursery (weed control, stump application and chemical thinning). Apples, pears, plums, cherries, damsons. Amenity vegetation.
Maximum individual dose: } Full details are given in the Statutory Area on the attached leaflet
Maximum number of treatments: } (Crop Specific Information – marked #)
Latest time of application: }
Other specific restrictions: }

IMPORTANT: Note that goods treated under the terms of this Great Britain authorisation can only be legally marketed in Northern Ireland if they are in accordance with the statutory EU Maximum Residue Level (MRL) set under Regulation (EC) No 396/2005. This may also apply to residues in animal products where treated crops are fed to livestock. Growers are advised to draw this to the attention of distributors and retailers so that EU MRL breaches and any associated enforcement against goods marketed in Northern Ireland are avoided.

READ THE LABEL BEFORE USE. USING THIS PRODUCT IN A MANNER THAT IS INCONSISTENT WITH THE LABEL MAY BE AN OFFENCE.
FOLLOW THE CODE OF PRACTICE FOR USING PLANT PROTECTION PRODUCTS.

SAFETY PRECAUTIONS

Operator protection

- Engineering control of operator exposure must be used where reasonably practicable in addition to the following personal protective equipment:
- WEAR SUITABLE PROTECTIVE GLOVES when handling the concentrate and when handling contaminated surfaces.
 - WEAR SUITABLE PROTECTIVE CLOTHING (COVERALLS), SUITABLE PROTECTIVE GLOVES AND RUBBER BOOTS when using hand-held sprayers, hand-held rotary atomisers, weed wiper equipment, spot gun equipment or when making out stump treatments OR WHEN USING STEM INJECTION EQUIPMENT.
 - However, engineering controls may replace personal protective equipment if a COSHH assessment shows they provide an equal or higher standard of protection.
 - WASH HANDS AND EXPOSED SKIN before eating and drinking and after work.
- Recommendations apply to the use of this herbicide for the control of weeds growing in or by water and must be read in conjunction with the Official Code of Practice entitled "Guidelines for the Use of Herbicides on Weeds in or near Watercourses and Lakes" obtainable from Department of Environment and Rural Affairs (DEFRA publications 08459 556000), Scottish Executive, Environment and Rural Affairs Department, Department of Agriculture and Rural Development for Northern Ireland and the National Assembly for Wales Agriculture Department.
- The Water Act, 1989, The Water Resources Act 1991, the Control of Pollution Act 1974, The Northern Ireland Water Resources Act 1992 and the Control of Pollution and Local Government (Northern Ireland) Order 1978, may apply to the act of applying Roundup ProVantage for the control of weeds growing in or by reservoirs and water courses, e.g. rivers, streams, ditches, drains and ponds/lakes discharging into such water courses.

Storage and disposal:
KEEP AWAY FROM FOOD, DRINK AND ANIMAL FEEDINGSTUFFS. KEEP OUT OF REACH OF CHILDREN. KEEP IN ORIGINAL CONTAINER, tightly closed, in a safe place. RINSE CONTAINER THOROUGHLY by using an integrated pressure-rinsing device or manually rinse three times. Add washings to sprayer at time of filling and dispose of safely. Triple rinsed containers may be disposed of as non-hazardous waste.

MAPP: 21197

RECOMMENDATION TABLES

AREA OF USE	CROP/ SITUATION	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
ALL EDIBLE AND NON-EDIBLE CROPS (DESTRUCTION, BEFORE SOWING/PLANTING)	Vegetation management	-	Annual weeds	1.0	80-250 L/ha* or hand-held equipment section	*Rotary atomisers may be used at a water volume of 10-40 L/ha. Ensure droplet diameter falls within the range 200-300 microns Do not use in or alongside hedgerows
			Perennial grasses	3.0		
			Perennial broad-leaved weeds	3.75		
GRASSLAND - DESTRUCTION	GRASS	Short rotation Ryegrass, longer leys and permanent pasture	Short rotation Ryegrass with annual weeds	2.25	150-250 L/ha*	Treat EITHER before grazing/mowing in June-Oct, when growth is 30-60 cm, not dense and lacking mature seeds, OR regrowth after grazing/mowing. Select the application rate which controls/destroys the least susceptible weed and grass species present in the sward. Grass may be conserved or grazed by cattle, dairy cows or sheep 5+ days after spraying. REMOVE POISONOUS PLANTS BEFORE GRAZING/MOWING. If Ragwort is present, the guidance in the DIRECTIONS FOR USE must be followed. ONLY direct drill grass and clover EITHER into 1-2 year leys without mat, 5+ days after spraying, OR long leys with some mat, in the spring following autumn application.
			Leys 2-4 years old with perennial grass weeds	3.0		
			Long leys 4-7 years old with perennial broad-leaved weeds	3.75		
			Permanent pasture See Weed Table below	4.5		

APPLICATION RATE FOR GRASSLAND DESTRUCTION					
2.25 L/HA		3.0 L/HA		3.75 L/HA	
Annual Meadow-grass Common Chickweed Common Mouse-ear Dock Seedlings Italian Rye-grass Mayweed species	Meadow Fescue Meadow Foxtail Rough Meadow-grass Speedwell species Timothy	Black-bent Broad-leaved Dock Cock's-foot Common Bent Common Couch Creeping Bent	Creeping Soft-grass Curled Dock Perennial Rye-grass Plantains Soft Brome Yorkshire Fog	Bracken** Common Sorrel Common Nettle Creeping Buttercup* Creeping Thistle Daisy Dwarf Thistle Perennial Sow-thistle	Red Clover Sedges Sheep's Sorrel Soft Rush Spear Thistle Tufted Hairgrass Yarrow
				Common Ragwort Hard Rush Heath Rush Jointed Rush <i>Molinia</i> (Purple Moor-grass)	<i>Nardus</i> (Mat grass) Red Fescue White Clover* Yellow Rattle Sheep's Fescue

* White clover is best cut in June and sprayed one month later ** At full frond expansion

AREA OF USE	CROP/SITUATION	TARGET WEEDS/ USAGE	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
NATURAL SURFACES NOT INTENDED TO BEAR VEGETATION, PERMEABLE SURFACES OVERLYING SOIL, RAILWAY BALLAST	Vegetation management	Including roadsides, paths and along fences	Annual weeds	1.0	Hydraulic sprayers (boom and knapsack) at water volumes 80-250 L/ha or Rotary atomisers* or hand-held equipment. See Mixing & Spraying section.	Do not use under polythene or glass.
			Perennial grasses and broad-leaved weeds	3.75		
HARD SURFACES (EXCLUDING RAILWAY BALLAST)	Vegetation management	including roadsides, paths, concrete and alongside walls	Annual weeds	1.0	Hydraulic sprayers (boom and knapsack) at water volumes 80-250 L/ha or Rotary atomisers* or hand-held equipment. See Mixing & Spraying section.	Apply this product carefully. Ensure spraying takes place only when weeds are actively growing (normally March to October) and is confined only to visible weeds including those in the 30 cm swath covering the kerb edge and road gully – do not overspray drains
			Perennial grasses and broad-leaved weeds	3.75		
AMENITY VEGETATION	Vegetation management	Ornamental areas	Annual weeds	1.0	Hydraulic sprayers (boom and knapsack) at water volumes 80-400 L/ha or Rotary atomisers*at water volumes 10-40 L/ha or hand-held equipment. See Mixing & Spraying section.	Do not use under polythene or glass.
			Perennial grasses and broad-leaved weeds	3.75		

* Rotary atomisers may be used at a water volume of 10–40 L/ha. Ensure droplet diameter falls within the range 200-300 microns

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Japanese Knotweed control

Japanese Knotweed is an invasive alien species reducing biodiversity in areas where it become established and propagating from tiny fragments of root, often spreading along watercourses. It is scheduled under the Wildlife and Countryside Act 1981 and all parts of the plant must be treated as Controlled Waste under the Environmental Protection Act 1990.

Roundup ProVantage can be used alone as part of an eradication programme or as part of an integrated programme in conjunction with soil disturbance or removal. Dormant rhizomes will not be controlled by Roundup ProVantage, but may be stimulated to grow by soil disturbance and then sprayed. It is particularly suitable for use near water. Sites must be monitored for at least three years and re-treated as necessary.

AREA OF USE	TARGET WEED	METHOD	APPLICATION RATE / WATER VOLUME	APPLICATION TIMING AND GUIDANCE
Amenity vegetation; Forestry; Natural surfaces not intended to bear vegetation, permeable surfaces overlying soil, hard surfaces; Enclosed waters, open waters, land immediately adjacent to aquatic area	Japanese Knotweed	Foliar application	3.75 L/ha Hydraulic sprayers 80-250 L/ha or hand-held equipment	For best results apply after the onset of flowering (usually August/September) but before dieback. Use specialist extending hand lances for stands 2-3 m tall. Good coverage is essential; spray the underside as well as the upper surface of the leaves. Or Only as part of an integrated programme, spray when stems are 1-1.5 m high (Usually at end of May) and repeat once re-growth reaches 1-1.5 m again later in same season or the following year. (More re-growth will occur from this timing.)
		Stem filling technique	10 ml of 15% solution per stem	See National Trust Methodology for full details*. Cut stems approx. 200mm above base of cane & 40mm above node. Rupture the central stem tissue with a screwdriver and use a spot gun to insert Roundup ProVantage into the hollow stem within 15 minutes of cutting.
		Stem injection technique	1.5 ml of neat solution per stem	Using specialist stem injection equipment, inject directly into individual stems.
		Hand-held weed wiper	1 part Roundup ProVantage to 3 parts water	Use where overall spraying is not desirable or target plants are small or unsuitable for stem filling, (< 8 mm) e.g. in re-treatment following foliar spraying.

AREA OF USE	CROP/SITUATION	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
ORCHARDS	APPLE, PEAR, PLUM, CHERRY, DAMSON	Perennial grasses, broad-leaved weeds	All levels of most species	3.75	200-400 L/ha	Spray AFTER autumn leaf-fall and BEFORE: Apples, pears - green cluster stage Stone fruit - white bud stage Treat root suckers in late spring only. Trees must have been established 2+ years. Avoid contact with tree 30+ cm above ground.
		Root suckers	-	3.75		
ENCLOSED WATERS, OPEN WATERS, LAND IMMEDIATELY ADJACENT TO AQUATIC AREAS	-	Emerged Weeds - Reeds, Rushes, Sedges, Grasses and Watercress	All levels/species	3.75	200-400 L/ha Optimum 250 L/ha or hand-held equipment (p.12)	Consult appropriate Environment Agency regional office before use. On water-lilies it is preferable to use a tractor or boat-mounted sprayer. During spraying do not exceed a pressure of 2.0 bars (30 p.s.i.). When using a tractor mounted sprayer do not exceed 8 kph (5 mph). Use boat-mounted sprayers at slowest practical speed. Always apply against direction of any current. When disturbed by wash, WATER-LILIES may require re-treatment.
		Floating Weeds - White water-lily - Yellow water-lily	All levels	4.5	100-200 L/ha or hand-held equipment (p.12)	

AREA OF USE	CROP/SITUATION	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
GRASSLAND, PLANT FREE AREAS, ORNAMENTAL PLANTINGS, AMENITY VEGETATION	-	Individual weeds	All levels	1:3 dilution with water for wick-type weed wipers OR 1:2 dilution with water in hot, dry conditions. For 'new generation' wipers consult the manufacturer for guidance.		Weeds must be 10+ cm taller, and wiper 5+ cm higher, than desired vegetation. Contact Bayer or your distributor for specific recommended weed wiper applicators. Treated POISONOUS WEEDS must be removed or allowed to completely degenerate before grazing or conserving. If Ragwort is present, the guidance in the 'DIRECTIONS FOR USE' must be followed.

Forestry weed control

Roundup ProVantage can be used for site preparation and for weed control in planted out trees.

AREA OF USE	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA.	WATER VOLUME	APPLICATION GUIDANCE
Forestry: - Pre-planting	Arable land, planting, replanting, & grassland areas	Arable weeds Grassland weeds	3.0 3.75	Hydraulic sprayers: 80-250 L/ha or rotary atomizers: 40 L/ha*	All tree species may be planted 7 days or more after treatment *Where rotary atomisers are used their droplet diameter must fall within 200-300µm.
Forestry: - Post-planting (directed) in conifers & broad-leaved trees	Clean-up around trees with knapsack applications.	Annual/perennial grasses and broad-leaves	3.0	Knapsack sprayers: 200-250 L/ha or Weed wiper mini: 1 part ROUNDUP PROVANTAGE to 3 parts water	It is ESSENTIAL to use a TREE GUARD for all applications made in the growing season. Treat bracken after frond tips are unfurled but before senescence. Treat heather late August to end September. All other woody weeds are treated June-August, before leaf senescence (but after new growth of crop has hardened).
		Woody weeds: Bracken/Beech Brush/Brambles Sycamore/Oak Hazel/Willow/Ash	2.25		
		Heather (peat soils) Heather (mineral soils)	3.0 4.5		
		Rhododendron (*)	7.5		
				250 L/ha	Cut back and treat regrowth when at least 1 metre in height throughout the entire coppice.

(*) For improved control of Rhododendron add Mixture B (ADJ AO570) at a concentration of 2% final water volume to 6.0 L/ha of Roundup ProVantage. Application using the weed wiper is not suitable.

AREA OF USE	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/HA.	WATER VOLUME	APPLICATION GUIDANCE
Forestry: - Post-planting (overall dormant season in certain conifers – conifer release)	Grass weeds - Lowland areas - Upland areas	Black Bent, Cock's-foot, Common Couch, Creeping Soft-grass, False Oat-grass, Fescues, Meadow-grasses, Other Bent species, Purple Moor-grass, Sweet Vernal-grass, Tufted Hair-grass, Wavy Hair-grass, Wood Small-reed (Bush grass)	1.0 1.5	Hydraulic sprayers: 200-250 L/ha or Hand-held equipment - see 'Mixing and Spraying' section	Species safe to spray when fully dormant and leader growth has hardened: Corsican, Lodgepole and Scots Pines, Norway Spruce, Sitka Spruce, Lawson Cypress, Western Red Cedar. Douglas Fir and Noble Fir – safe to spray when fully dormant and leader growth has hardened but NOT in spring. If overall application takes place after the optimum timing weed control may be reduced. It is advisable to spray a limited area of forest to test crop safety under local conditions before widespread overall application in subsequent years. These recommended application rates refer to forestry usage only. Inadequate control may result if used in other areas. Caution: The timing of hardening of leader growth varies considerably between locations and between seasons. It may occur as early as the end of July or be delayed to October or later. To avoid damage to Lammas growth, sprays should be directed away from leaders.
			1.5 1.5 2.25		
	Bracken Beech & Birch Brambles	All levels of all species All levels of all species All levels of all species			
Forestry: - Stump application for chemical thinning	Deciduous trees Coniferous trees	All species All species	7.5% solution of Roundup ProVantage in water 15% solution of Roundup ProVantage in water		Apply the solution to saturate the rim of the newly cut surface, with a suitable adapted clearing saw, spot gun or paintbrush. Treat as soon as possible after felling, in the period November to March/April. Do not apply in the period of active sap flow in the spring/early summer. Do not cut trenches or drill holes and fill with the solution or use undiluted product. Note: for ease of identification of treated areas a suitable, commercially available, water-soluble dye may be added to the prepared spray solution.
Forestry: - Chemical thinning by injection of tree stems	Coniferous and deciduous species	-	1.5 ml neat Roundup ProVantage per cut per 10 cm diameter (or less) tree		Use a hatchet to cut one notch in trees up to 10 cm diameter and apply 1.5 ml of the solution to each cut. Use two or three notches in trees over 10 cm diameter. Do not treat in the period of active sap flow in the spring/early summer.